Apex Basket Crane 2

Current application behavior

Code analysis dated September 25 2026

The application controls basket crane 2 workflows, maintains basket locations, and coordinates exports, imports, conveyors, and destacker operations. This document lists its current behavior, followed by code defects that affect those workflows.

Scope: the checked-in apexBasketCrane2 solution and its related source. This is static analysis; the application was not run and no equipment or databases were connected. PLC commands describe requests made by the application, not verified physical machine actions.

1 Startup and connections

- Prevents another instance with the same executable filename from starting.
- Shows a splash screen and opens maximized.
- Connects to crane 2, the old oven, and the new oven.
- Opens the basket database and EPICS database; production startup fails if either connection cannot open.
- Uses connection settings compiled into the application.
- Creates missing application log, mission, and export tables.
- Trims each application log table to its latest 10,000 records at startup.
- Loads stored basket positions and checks for unfinished missions.

2 Main display

- Displays oven positions, buffers, return positions, conveyors, destacker positions, and crane carriage.
- Shows basket numbers, contents, destination classifications, export assignments, and locks.
- Highlights mission source and destination positions.
- Displays crane coordinates, targets, current mission, next mission, and PLC step.
- Provides mouse-wheel zoom, middle-button pan, and Zoom Extents.
- Shows detailed basket/profile information in tooltips.
- Updates the window title with the current time.

3 Operating modes

- Starts in Missions mode.
- Provides Missions, Exports, and Modify selection modes.
- Provides separate export lists for HCB/SB, anodizing, packing destacker 1, and packing destacker 2.
- Selecting the automatic-controls page switches layout selection back to Missions mode.
- Automatic scheduling options start disabled.

4 Manual crane missions

- Clicking an eligible occupied position selects the source.
- Clicking an eligible empty position attempts the move immediately.
- Clicking the selected source or another occupied position cancels source selection.
- Escape cancels a pending selection.
- Rejects moves when the crane is unavailable, communications report an error, the source is empty, or the destination is occupied.
- Rejects locked, obstructed, unsupported, or already-reserved positions.

- Persists accepted missions and sends their endpoints to the PLC.

5 Stack and obstruction rules

- Prevents moving a basket with another basket immediately above it.
- Prevents using an upper level when the level immediately below is empty.
- Checks configured obstructions from adjacent oven columns.
- Reserves entire position columns against conflicting missions.
- Excludes old-oven columns 660 and 680 from normal manual source/destination selection.
- Represents three usable levels in each buffer and three return-stack levels.

6 Mission execution and tracking

- Maintains one next mission and one active mission.
- Promotes the next mission when the PLC reports starting.
- Moves the recorded basket from source to carriage when the PLC reports pickup.
- Moves it from carriage to destination when the PLC reports completion.
- Acknowledges mission stages to the PLC.
- Clears completed mission records.
- Logs mismatches such as an occupied destination or missing carriage basket.
- Checks mission status every second.

7 Automatic scheduling

- Supports independent enable switches for four export categories, north imports, south imports, and EPICS queue population.
- Evaluates missions every 2.5 seconds and immediately when a mode is enabled.
- Considers the first basket in each enabled export queue.
- Uses this fixed priority: HCB/SB, anodizing, packing 1, packing 2, north import, then south import.
- Sends all export categories to the north conveyor handoff.
- Imports north/south baskets into the return stack.
- Checks route occupancy and downstream packing workload.
- Does not automatically try a lower-priority candidate when the selected candidate fails final mission validation.

8 Automatic uncovering

- Detects baskets blocking a queued export basket.
- Attempts to relocate a blocker into an available north or south buffer position.
- Searches buffer levels from lower to higher.
- Moves a blocker already in one buffer only to the other buffer.
- Leaves the exported basket queued while relocating blockers.

9 EPICS basket information

- Reads profile, quantity, length, temper, sales order, order line, finish code, standard practice, customer, next department, and production stop time.
- Loads contents for basket/rack numbers 1–130 with nonzero quantities.
- Converts order length to millimetres.
- Classifies contents using department, finish, customer, and standard-practice text.
- Classifies non-DESTACK contents as None.
- Routes matching finishing/customer/practice cases to HCB; remaining DESTACK cases become HCA.

- Uses the designated top contents to classify the basket.
- Can populate export queues from the highest occupied basket in each eligible oven/buffer column.
- Uses matching sales order and line to continue selected destacker workloads.

10 Export list management

- Lets operators assign baskets to one of four export queues.
- Manual reassignment removes the basket from its previous export queues.
- Appends new entries at the end of the queue.
- Lets operators move entries up/down or remove them.
- Shows queue counts and an empty-list message.
- Refreshes lists every five seconds.
- Manual packing assignment requires maximum profile length ≤7660 mm.
- Removes completed north-handoff baskets from crane-2 export queues.
- Adds qualifying packing baskets to the corresponding crane-3 queue.

11 Conveyor routing

- Writes a destination code based on export membership.
- Supports None, Anodizing, Destacker For SB, To Press, In from Destacker, Anodizing Overlength, and Packing Destacker.
- Contains routing thresholds of 7821 mm for HCB/anodizing and 7660 mm for packing.
- Logs destination writes to a local text file.
- Does not automatically clear an existing destination when the basket has no export-queue membership.

12 Returning baskets to crane 1

- Fills return positions from bottom to top.
- Requests return automatically after a completed delivery fills all three levels.
- Provides a toolbar action to request return earlier.
- Sends the bottom basket number to the old-oven PLC.
- Does not itself clear all three basket locations in that request function.

13 Basket and position correction

- Ctrl-click opens position information where supported.
- Shift+left-click reloads position data.
- Allows basket-number and position-lock editing.
- Uses basket number 0 to represent no basket.
- Shows basket contents and movement history.
- Blocks basket-number changes for positions participating in stored missions.
- Attempts to prevent declaring a basket already located elsewhere.
- Directs most PLC-controlled position edits to the conveyor interface.
- Conveyor editing writes basket number and destination directly to the PLC.

14 Crane HMI controls

- Provides Auto, Manual, Active Stop, and Reset Alarms.
- Provides X/Y jogging and Z up/down controls.
- Provides basket-pin and safety-pin lock/unlock.
- Displays position, target, velocity, frequency, current, proximity, and on-target feedback.
- Provides axis trend plots.

- Provides Continue, Abort, End, Clear Next, and Clear Active Database Mission.
- Provides encoder resets and an interlock-bypass command.
- F1 sends the Active Stop command.
- Basic HMI provides alarm reset and alarm/warning displays.

15 Conveyor HMI controls

- Displays eight conveyor positions and their proximity switches.
- Provides conveyor Auto, Manual, and alarm reset.
- Provides individual forward/backward jogging.
- Provides paired conveyor movement commands.
- Provides conveyor rotation, lift up/down, and side movement.
- Graphical movement buttons send a command on press and zero on release.
- Displays active conveyor commands and alarms.

16 Destacker HMI controls

- Displays health feedback for eight pincers.
- Provides individual pincer enable/disable commands.
- Provides a Spacers on Belts control.
- Displays destacker alarms.

17 Recovery access and logging

- Warns about a stored active mission at startup and offers to clear it.
- Attempts to restore a stored next mission.
- Attempts to return recorded basket locations to the source after abort.
- Password-protects the PLC communication dialog and unlocking selected advanced mission buttons.
- Initially disables Abort, End, and Clear Active Database controls whenever the full HMI opens.
- Opens the full HMI and ordinary position editor without a password.
- Records general, verbose, GUI, position, and configured PLC-message history.
- Shows recent logs and larger history dialogs.
- Consolidates consecutive identical GUI messages by updating their timestamp.

18 Current defects affecting behavior

- Settings password enforcement is broken: the result is discarded by a comma expression.
- The additional password check inside Clear Active Database Mission has the same defect, although its advanced-button unlock remains separate.
- A bConveyorState/BConveyorState mismatch prevents the intended handoff-readiness update through the shown path.
- Handoff routing can evaluate the destination's previous contents length before loading the arriving basket.
- An immediate export-marker helper incorrectly maps packing destacker 2.
- Automatic uncovering can run for an individually disabled export category when another automatic option is enabled.
- Maintenance mode is forced true in the conveyor update handler.
- Stored next missions can reject themselves during restart validation.
- Abort recovery may require another status-processing cycle before clearing.
- Duplicate-location detection misses baskets recorded in multiple locations.
- The crane drawing's Z value uses a different PLC index from the HMI.
- Profile top and bottom ordering is inconsistent between helpers.

- The current production EPICS classifier never returns anodizing, despite having an automatic anodizing branch.
- Failure to open the local destination log exits the application.
- Database transfers and queue updates use multiple calls without an explicit transaction in this application layer.

Inactive production workflows

Emulation, legacy crane functionality, and the unused generic connection-settings dialog are not active production workflows in the checked-in configuration.

Detailed source reference

The companion APP_BEHAVIOR_INVENTORY.md contains 182 detailed entries, source references, PLC command mappings, and 320 configured alarm and status labels.